

Nhut Minh Nguyen

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Summary and Objective

Nguyen Minh Nhut is an Artificial Intelligence undergraduate and Research Assistant at FPT University, specializing in multimodal emotion recognition and human-centered AI. He builds robust audio-text-vision systems using Transformer encoders, graph-based learning, and semi-supervised/federated paradigms to improve generalization under limited labels and privacy constraints, delivering reproducible pipelines and deployable inference workflows.

Education

FPT University (Ho Chi Minh City), B.Sc. in Artificial Intelligence Sep 2022 – May 2026
(Expected)

- **Focus:** Machine Learning, Deep Learning, Speech Processing, Computer Vision, Human-Computer Interaction.
- **Relevant Coursework:** Machine Learning, Deep Learning, Computer Vision, Speech Processing, Data Structures & Algorithms.
- **GPA:** 8.45.

Experience

Research Assistant, AiTA Lab – FPT University Mar 2023 – Present

- Research and develop multimodal emotion recognition systems with Transformer encoders, feature fusion, and cross-modal attention.
- Design graph learning modules to capture contextual dependencies for emotion recognition in conversation.
- Explore semi-supervised and federated learning paradigms for emotion recognition to improve robustness under limited labels and privacy constraints.
- Build end-to-end PyTorch pipelines for training and evaluation, and deploy inference workflows.
- Additional research interests: active learning, handwritten mathematical expression recognition, medical federated learning, and applied computer vision for real-world applications.

Founder & Lead, AIoT Research Pioneer Club (FARPC) – FPT University Sep 2025 – Present

- Lead a research-driven student community focused on AI/AIoT self-learning, research thinking, and academic dissemination.
- Organize weekly seminars, reading groups, and mentoring activities; support members in project execution and paper-style reporting.
- Coordinate collaboration with mentors/labs and curate learning materials for multimodal learning and applied machine learning pipelines.

AI Research Intern, Pythera AI Jan 2025 – Apr 2025

- Worked as an AI engineer to develop an AI-based image white balance solution, improving color consistency and visual quality under diverse lighting conditions.
- Participated in model compression to reduce latency and deployment footprint while maintaining accuracy.
- Supported production deployment by exporting models to ONNX and validating inference consistency across environments.

Grants

Multimodal Fusion in Speech Emotion Recognition — FPT University (Hanoi, Vietnam) Apr 2025 – Oct 2025

- Grant number: DHFPT/2025/10.

Selected Publications

Enhancing multimodal emotion recognition with dynamic fuzzy membership and attention fusion (*First author*) Feb 2026

Engineering Applications of Artificial Intelligence.

DOI: 10.1016/j.engappai.2025.113396. — Code: github.com/nhut-ngnn/FleSER.

- Proposes a multimodal emotion recognition approach with dynamic fuzzy membership with attention-based fusion.
- Targets robust affect modeling by adaptively weighting expressive cues across modalities.

Multimodal fusion in speech emotion recognition: A comprehensive review of methods and technologies (*First author*) Jan 2026

Engineering Applications of Artificial Intelligence.

DOI: 10.1016/j.engappai.2025.112624. — Repo: [awesome-multimodal-fusion-emotion-recognition](https://github.com/nhut-ngnn/awesome-multimodal-fusion-emotion-recognition).

- Provides a structured review of multimodal fusion techniques for speech emotion recognition across audio, text, and vision.
- Summarizes trends in architectures, features, datasets, and evaluation practices to guide future research and reproducibility.

CemoBAM: Advancing Multimodal Emotion Recognition through Heterogeneous Graph Networks and Cross-Modal Attention Mechanisms (*First author*) Sep 2025

APNOMS 2025 (Kaohsiung, Taiwan).

DOI: 10.23919/apnoms67058.2025.11181320. — Code: github.com/nhut-ngnn/CemoBAM.

- Introduces a heterogeneous graph-based framework with cross-modal attention for multimodal emotion recognition.
- Models relational structure across modalities to capture interaction patterns beyond sequential fusion.

HemoGAT: Heterogeneous Multimodal Speech Emotion Recognition with Cross-Modal Transformer and Graph Attention Network (*First author*) 2025

Advances in Electrical and Electronic Engineering. To appear.

- Proposes a heterogeneous multimodal emotion recognition approach using a cross-modal Transformer and graph attention for structured fusion.
- Targets improved modeling of modality-specific cues and their interactions for emotion prediction.

ALMUS: Enhancing Active Learning for Object Detection with Metric-Based Uncertainty Sampling (*Co-author*) Sep 2025

APNOMS 2025 (Kaohsiung, Taiwan).

DOI: 10.23919/apnoms67058.2025.11181447.

- Proposed ALMUS, a metric-based uncertainty sampling approach with dynamic class-wise budget allocation to balance classification and localization.
- Validated on benchmark datasets, showing improved performance and faster convergence compared with active learning baselines.

Awards

The 27th Euréka Award for Student Research — Consolation Prize Dec 2025

- Topic: Speech Emotion Recognition with Heterogeneous Graph Construction.

FPT Research Festival (ResFes 2025) — Consolation Prize Aug 2025

- Topic: Speech Emotion Recognition with Heterogeneous Graph Construction.

Student Research Competition, FPT University — Champion Prize Apr 2025

Technologies

Languages: Python, C/C++, Java, SQL, Bash.

Speech & Signal Processing: Speech-to-text (ASR), speaker/emotion analysis, audio preprocessing (resampling, normalization, VAD), time–frequency analysis (spectrogram/MFCC), feature extraction.

Frameworks/Libraries: PyTorch, Hugging Face Transformers, TorchAudio, NumPy, scikit-learn.

Tools: Git/GitHub, Linux, ONNX, Weights & Biases (wandb), LaTeX, Streamlit.